SkillForge

Requirement Analysis

1. Introduction

1.1 Purpose

This Requirement Analysis explains the requirements of the SkillForge prototype before system design and implementation. SkillForge is an interactive, puzzle-based skill development prototype. This document gives a practical view of the approved requirements and provides a base for DFD preparation, ER and database design, system design, Unity setup, implementation, and testing.

The analysis is based on the requirements already defined in the SkillForge Software Requirements Specification (SRS). It does not add new gameplay features, technologies, platforms, or design decisions outside the approved project scope.

1.2 Objectives of the Analysis

• Identify the functional requirements that the SkillForge prototype must fulfil.

• Identify the users and the interactions available to them.

• Identify the required user-interface and interaction behaviour.

• Identify the data and persistence requirements needed for gameplay and player progression.

• Identify the relevant system and non-functional requirements.

• Identify project constraints, assumptions, and dependencies that affect implementation.

• Provide a clear requirement baseline for system design, database design, implementation, and testing.

1.3 Basis of the Requirement Analysis

This Requirement Analysis is derived from the approved SkillForge SRS and its finalized project decisions. The SRS defines the prototype's purpose, scope, expected behaviour, major features, user interactions, interface requirements, and system requirements.

The SRS is therefore used as the main reference for this analysis. If a requirement is not defined in the SRS, it is not treated as a current SkillForge requirement unless the project scope is reviewed and approved.

2. Scope & User Analysis

2.1 Project Scope

SkillForge is an interactive puzzle-based skill development prototype designed as a self-contained project experience. It combines an ancient-futuristic environment, interactive interface elements, progression systems, and puzzle-based challenges around a central HUB and connected Challenge areas.

The prototype follows a controlled progression flow: Opening Sequence and Hall Awakening, Guest HUB interaction, Login/Sign-Up, Data Feed, Glyph Conversion, Forge progression, Challenge access, Challenge interaction, Portal activation, Reward presentation, and return to the HUB.

The current scope includes:

•

Opening Experience and Hall Awakening

•

HUB management and progression

•

Account interaction through Login/Sign-Up

•

Data Feed and Glyph Conversion

•

Forge progression and Forge Vision

•

Skill Tree progression

•

Challenge Selection and Challenge Chamber

•

Ring System and its analysis stages

•

Portal Activation and Reward Presentation

•

Progress and Save Management

•

Notifications, Hints, Settings, and other approved interface support • Return to HUB after the defined Challenge and Reward flow

These functions make up the controlled prototype loop and will be considered during design and implementation. The prototype is limited to the approved academic demonstration scope. It is not intended to be a complete commercial-scale game or an unrestricted free-roaming experience.

2.2 User Analysis

The Player is the primary user of SkillForge. The player interacts with the opening experience, Hall, HUB, interface panels, account flow, progression systems, Challenge experience, puzzle mechanisms, Portal, Reward flow, and return-to-HUB process.

Guest User

The Guest User represents the player before completing the account-related flow and receiving the required progression data. The Guest state allows the approved initial HUB interaction, while progression remains restricted until the required account flow is completed.

Authenticated User

The Authenticated User represents the same player after completing the Login/Sign-Up flow and receiving the required Data Feed. This state allows the player to continue through the subsequent systems.

Guest User and Authenticated User are not separate roles. They are two progression states of the same Player. No separate Admin, Teacher, Moderator, Multiplayer Player, or Content Creator role is defined for the current prototype.

2.3 Scope Boundary Relevant to Implementation

For the current implementation, SkillForge is a fixed-camera 2.5D Windows Desktop Unity application. The approved environment uses Unity 6.3 LTS and C#, with MainScene and ChallengeScene forming the two-scene structure.

The project is limited to the approved features and architecture. The SRS defines constraints such as the two-scene structure, fixed-camera 2.5D presentation, responsibility-based Manager/Script organization, and SaveManager as the sole persistent save handler.

3. Functional Requirement Analysis

The Functional Requirement Analysis identifies the main functions that the SkillForge prototype must provide. It summarizes the required behaviour while keeping the important conditions and outcomes defined in the SRS.

3.1 Opening Sequence

The application will show the defined cinematic Opening Sequence when it starts. The sequence will move through loading, connection failure, darkness, forge sound, connection-established, crystal, narrative, awakening, logo, and Hall-reveal stages. It ends with the “Press ENTER to Begin” interaction before the player moves to the Hall/HUB flow.

3.2 Guest HUB

After Hall Awakening, the system enters the Guest HUB. The player can view the Guest-mode interface, while progression-related interactions remain restricted until the required account flow is completed. The Profile Panel provides access to Login/Sign-Up, and the defined panel and ESC behaviour is maintained.

3.3 Login / Sign-Up

The Profile Panel provides Login and Sign-Up. After successful account interaction, the required player data is initialized or loaded and the player can continue toward the Active HUB. An unsuccessful login does not move the player into the Active progression state.

3.4 Data Feed

After account initialization, the system provides the required player and progression data to the relevant HUB interfaces. This data is made available to the appropriate progression systems before the player continues through the normal Active HUB flow.

3.5 Glyph Conversion

Relevant HUB information and progression elements change into their defined Glyph-based state. The player can expand a Glyph to open its associated panel and return it to Glyph form using the defined Back/ESC interaction.

3.6 Forge Core

Forge Core provides the initial progression path inside the HUB: Forge Core → Obtain Forge Vision → Skill Tree Update → Forge Core returns to Glyph. The resulting progression state is maintained according to the defined progression and save behaviour.

3.7 Forge Vision

Forge Vision provides an analysis mode during Challenge gameplay. It allows the player to observe Formation Status, Alignment, Energy Flow, and Core Resonance without directly showing the puzzle solution. The analysis updates according to the current Challenge condition.

3.8 Skill Tree

The Skill Tree provides the progression interface for available abilities. It remains part of the HUB and reflects the player's current progression state.

3.9 Challenge Selection

After the required progression, the Challenge Portal provides access to Challenge Selection. The selected Challenge moves to Level Selection and then enters the Challenge through the defined Portal Warp and transition flow.

3.10 Challenge Chamber

The Challenge Chamber is the main puzzle environment. It uses the fixed-camera presentation and provides the Challenge Formation, Ring System, Forge Vision, and analysis interfaces. The Challenge begins in its defined unsolved state and is completed when the required Alignment, Energy Flow, and Core Resonance conditions become stable.

3.11 Ring System

The Ring System is the main interactive puzzle mechanism. It has three connected rings: Middle Ring — Physical Alignment; Outer Ring — Energy Flow; Inner Ring — Core Synchronization. The rings start in a Broken Formation state and respond to player interaction in real time. When Alignment, Energy Flow, and Core Resonance become stable, the Formation becomes Stable and the Challenge Portal is activated. Forge Vision provides analysis without directly showing the solution.

Scope Note

The exact rotation degrees, rotation counts, sensitivity values, and solution positions are not defined in the approved SRS. These values should therefore not be invented during implementation.

3.12 Alignment Analysis

When Forge Vision is active, the system provides Alignment Accuracy, Structural Offset, Alignment Stability, and Structural Response. The information reflects the current puzzle state without directly showing the required solution.

3.13 Energy Flow

When Forge Vision is active, the system provides Energy Sources, Energy Flow, Energy Leakage, and Channel Stability. These values update according to the current puzzle state without directly showing the required solution.

3.14 Core Resonance

When Forge Vision is active, the system provides Resonance Level, Core State, Connection Quality, and Resonance Stability. The information reflects the current synchronization state.

3.15 Portal Ignition

The Challenge Portal becomes active after the required Alignment, Energy Flow, and Core Resonance states are stable and the Formation reaches its completed state. Portal interaction starts the defined completion and transition flow.

3.16 Reward System

After successful Challenge completion, the Reward flow is presented. The earned reward is added to the player's progression, followed by the defined return-to-HUB flow. A separate Reward Chamber is not part of the current prototype.

3.17 Save System

The Save System maintains persistent player progression, completed Challenge state, rewards, and relevant settings. SaveManager remains the sole persistent save handler. Completed Challenge progress and rewards are saved, while incomplete Challenge progress is not retained.

3.18 Notification System

The system provides notifications for meaningful approved events such as Challenge completion, rewards, progression, and relevant system or error states. Notifications follow the defined queue and presentation behaviour without replacing the actual gameplay or completion state.

3.19 Settings

The Settings system provides the approved audio and interface settings. It remains accessible through the defined interface and follows the approved Apply, ESC, and persistence behaviour.

3.20 Return to HUB

After Challenge completion, Reward flow, or manual Challenge exit, the player returns to the HUB through the defined transition. Completed Challenge progression and rewards are updated and saved. If an incomplete Challenge is exited manually, its incomplete progress is not saved as completed.

4. UI & Interaction

4.1 HUB Interface

The HUB is the main interface of SkillForge and provides access to the main progression and information systems.

The player first enters the Guest HUB after Hall Awakening. Progression-related actions remain restricted until Login/Sign-Up is completed.

The main HUB areas are:

- Profile
- Forge
- Skill Tree
- Challenge
- Knowledge / Information
- Settings

The HUB uses the defined panel and Glyph interaction. A player can open an available information element into its panel and use Back/ESC to return to the previous state.

4.2 Account and Progression Interaction

The Profile Panel provides Login/Sign-Up. After successful account interaction, the required player data is loaded and the player can continue toward the Active HUB. An unsuccessful login does not move the player into the Active progression state.

The HUB then shows the player's progression through systems such as Forge Core, Forge Vision, and Skill Tree. The Skill Tree remains part of the HUB and does not open as a separate scene.

4.3 Challenge Interface

After the required progression, the player can select a Challenge through the Challenge Portal. The Challenge Chamber provides the main puzzle interface using the fixed-camera 2.5D presentation. It provides access to the Challenge Formation, Ring System, Forge Vision, and analysis information.

The player interacts with the Ring System to solve the Challenge. The interface shows the current Challenge state through the analysis information without directly showing the solution. The Challenge is completed when Alignment, Energy Flow, and Core Resonance become stable and the Formation reaches its completed state.

4.4 Forge Vision and Analysis Panels

Forge Vision works as an analysis mode during the Challenge. When it is active, the player can view the current Challenge state through Alignment Analysis, Energy Flow, Core Resonance, and Formation Status.

Alignment Analysis shows Alignment Accuracy, Structural Offset, Alignment Stability, and Structural Response. Energy Flow shows Energy Sources, Energy Flow, Energy Leakage, and Channel Stability. Core Resonance shows Resonance Level, Core State, Connection Quality, and Resonance Stability. These values change with the current Challenge state and help the player understand the puzzle without directly showing the solution.

4.5 Notifications, Hints and Settings

The system provides notifications for events such as Challenge completion, rewards, progression, and relevant system or error states. Hints guide the player during approved interactions without directly giving the puzzle solution. Settings provide the approved audio and interface options and follow the defined Apply, ESC, and save behaviour.

4.6 Portal, Reward and Return Interaction

When the required Challenge conditions become stable, the Challenge Portal becomes active. The player can interact with it to start the completion flow. After successful completion, the Reward flow is shown and the earned reward is added to the player's progression. The player then returns to the HUB and completed progress and rewards are saved.

If the player leaves an incomplete Challenge manually, it is not marked as completed and no completed-Challenge reward is given.

4.7 Overall Screen Flow

Main UI flow: Opening → Guest HUB → Login / Sign-Up → Active HUB → Challenge Selection → Challenge Chamber → Portal Activation → Reward Flow → HUB

Account flow: Guest HUB → Profile → Login / Sign-Up → Account Setup → Active HUB

Challenge flow: Challenge Selection → Challenge Chamber → Ring System → Stable Formation → Portal Activation → Reward → Return to HUB

All UI and interactions remain within the defined project scope. No additional screens, game areas, or interaction systems are added without approval.

5. Data & Persistence

5.1 Player Data

SkillForge needs basic player data to keep track of progress. The main player data includes:

- Username
- Player ID
- Level
- XP
-

Inventory

-

Unlocked skills and abilities

-

Challenge completion status

-

Reward and progression status

The player can be in Guest User or Authenticated User state. A new account starts at Level 1 with 0 XP, an empty inventory, and no completed Challenges. Forge Vision is locked at the start.

5.2 Runtime Data

During gameplay, current player data is handled by PlayerManager. It manages data such as XP, Level, Inventory, and Skill-Abilities while the game is running. Different game systems use this data according to the player's current progress.

5.3 Save Data

Important player progress must be saved so that it is available after the game is closed. SaveManager is responsible for saving and loading the approved player data and also handles auto-save and persistent data. The main saved data includes player identity, Level, XP, Inventory, unlocked abilities, and completed Challenge progress.

5.4 Challenge and Reward Data

Challenge progress is saved only after a valid Challenge completion. If a player leaves an incomplete Challenge, it is not counted as completed and no reward is given for that attempt. After successful completion, the reward and related progress are saved as part of the player's progression.

5.5 Data Flow

Player Action → Game System → Player Data Update → SaveManager → Saved Progress

Runtime data is handled during gameplay, while SaveManager handles the data that must remain available for the next game session.

The exact database tables, fields, and storage format are not defined at this stage. They will be decided during the database design work.

6. System & Non-Functional Requirements

6.1 Usability

SkillForge should be easy to understand and use. The player should be able to understand what can be done on each screen and panel, and the same type of action should behave consistently across the game.

The player should be able to move through the main flow from Guest HUB to Active HUB, Challenge, Reward Popup, and back to the HUB. Notifications, hints, analysis panels, and audio should provide useful feedback. Wrong or invalid actions should be stopped or handled correctly.

6.2 Reliability

The main parts of the game should work correctly during normal use. The Challenge should handle player actions, completion, and Portal movement correctly. When a Challenge is completed, its reward and progress should be saved correctly.

If the player leaves before completing a Challenge, it should not be marked as completed and no reward should be given. Invalid actions should keep the game in the correct state without unwanted changes.

6.3 Security

The game should protect player account and saved progress within the limits of this academic project. Guest User and Authenticated User should have the correct access. Guest Users should not receive account-based progression access before completing account setup.

Player progress and saved data are handled through SaveManager. Rewards and progress are updated only after valid Challenge completion and the save process.

Because SkillForge is an academic prototype, advanced commercial security systems such as server security or complex encryption are not required.

6.4 Maintainability

The project should have a clear structure so it is easy to change and maintain during development. Each Manager should have its own responsibility, and two Managers should not control the same system or data.

Scripts should also have clear responsibilities. PlayerManager handles player data during gameplay, while SaveManager handles saved data. Reusable game objects should use the approved Prefab system where needed.

6.5 Compatibility

The SkillForge prototype will run on the approved platform and setup. The target platform is Windows Desktop, and the project will use the approved Unity environment.

The prototype uses two approved scenes: MainScene and ChallengeScene. The Challenge uses a scene change when the player enters or returns from it. Other normal UI actions remain in the Main Scene.

Keyboard and mouse controls should work with the approved gameplay and UI, and all approved systems should work together without causing conflicts.

7. Constraints / Assumptions / Dependencies

7.1 Constraints

SkillForge is an academic game prototype, so development will remain within the approved scope and available resources.

The project will use the approved Unity setup, Windows Desktop as the target platform, and the two-scene structure of MainScene and ChallengeScene.

The gameplay flow, Challenge rules, Ring System, Forge Vision, reward flow, and save system will follow the defined requirements. New gameplay systems or features will not be added without a change to the approved scope.

The project will also use the fixed-camera 2.5D Challenge setup and the approved keyboard and mouse controls.

7.2 Assumptions

- The player will use a Windows Desktop system.
- The required Unity environment will be available during development.
- The player will use keyboard and mouse controls.
- The player will follow the defined game flow and Challenge rules.
- Required game assets and project files will be available when needed.
- Save and load functions will work through SaveManager.
- The approved game systems will work together as planned.

7.3 Dependencies

SkillForge depends on several systems working correctly together. PlayerManager is needed for player data during gameplay, while SaveManager is needed for saving and loading player progress.

The Challenge depends on the Ring System, Forge Vision, analysis panels, Portal system, and Reward flow. The HUB depends on the login/account flow and the player's current progression state.

The final game flow also depends on correct scene changes between MainScene and ChallengeScene. All these systems need to work together for the defined SkillForge gameplay flow to work correctly.

8. Traceability & Verification

8.1 Requirement Traceability

Each major SkillForge requirement should be connected to the part of the project where it will be used. The main connection is:

Requirement → System Function → Design → Implementation → Testing

For example, the Challenge requirements are connected with the Challenge system, Ring System, Forge Vision, Portal, Reward, and Save System. UI requirements are connected with the related screens, panels, popups, and interactions. Data requirements will be used later for data and database design.

8.2 Requirement Verification

Each requirement should be checked during development to make sure the final system works as planned. The following points should be verified:

•

The player can follow the defined game flow.

•

The HUB and Challenge interfaces work as planned.

•

Challenge actions give the correct result. •

Forge Vision shows the defined analysis information.

•

Completed Challenges give the correct reward.

•

Incomplete Challenges do not give completion progress or reward.

•

Player progress can be saved and loaded correctly.

•

Notifications, hints, and settings work according to the defined requirements.

8.3 Link with Further Development

This Requirement Analysis will be used as a reference for the next documentation and development work. The identified system flow and data will help in preparing the DFD, ER Diagram, and Database Design.

The requirements will also be used during System Design, Unity implementation, and testing. Any requirement that is not supported by the SRS should not be added during implementation without first reviewing the project scope.